United International University
Department of Computer Science and Engineering

CSE 3421: Software Engineering   Mid: Fall 2025

Total Marks: 30   Time: 1 hour and 30 minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all five questions. The numbers on the right of the questions denote their marks.

1. (a) **Write GIT commands** for the following **tasks**: [CO2] (6)

   I Create a new local GIT repository on your machine and link it to: https://www.github.com/dev/PROJECT-X.git

   II Create 2 new files named Alpha.txt and Beta.txt, and save a version of the files after staging them separately.

   III Create 2 new branches named Eagle and Falcon.

   IV On the Falcon branch, save the recent edit of the Beta.txt file, and on the Eagle branch, save the recent edit of the Alpha.txt file.

   V You forgot to set your username and email. Set your username to DevMaster and your email to dev@example.com globally.

   VI Someone modified the remote repository. Switch to the master branch and get the latest changes of the project to your machine. Merge all the work and branches, and upload everything to GitHub.

2. (a) What is the **difference** between Requirement **Verification** and Requirement **Validation**? Explain with proper **reasoning**. [CO1] (2)

   (b) You are managing the online system of a university. The portal is facing the following issues: (4)
   Case 1: Several lecturers receive an email asking them to click a link to "verify their portal account," which leads to a fake login page.
   Case 2: Students discover hidden keystroke-recording software on some lab computers.
   Case 3: An attacker phones administrative staff pretending to be an IT officer and convinces them to share their portal password "to fix a system error."
   Case 4: The university's attendance software suddenly deletes the full database on the first day of January without any manual action.
   What type of **cyber-attacks** are responsible for each case described above? Explain your answers with proper **reasoning**. [CO2]

3. (a) What is the **difference** between **Techincal** documentation and **Architectural** documentation? Explain with a suitable **example** for each. [CO1] (2)

   (b) A fintech startup is hired by a client to develop a mobile expense management app that tracks daily spending, categorizes expenses automatically, generates visual charts, and sends personalized notifications. The development team quickly designs and implements the full app based on the client's initial requirements, and delivers it after several weeks. When the client tests it, they notice multiple issues: the charts track income instead of expenses, notifications are sent weekly instead of daily, and the automatic categorization often mislabels transactions. Additionally, the client requests new features like budgeting goals and bill reminders, which were not considered in the original design. The client requires quick regular updates and proper monitoring for the project. (4)
   Which software development **model** would have been most suitable in this scenario? **Explain** your reasoning clearly. [CO2]

4. (a) Why is Spiral model called a **Meta-model**? Explain with proper **reasoning**. [CO2] (2)

   (b) A team is building a smart parking app. To help new developers understand the system, they describe it as a parking lot with entry gates, sensors, and cars. So everyone can visualize how parts interact. Developers merge their code into the main branch several times a day, and automated tests immediately run to catch issues. Occasionally, they notice that some code is confusing and reorganize it to make it cleaner and easier to maintain. Throughout the project, a representative from the client works closely with the team every day, answering questions and providing feedback on features as they are built. (4)
   Which **XP practices** are being applied in this scenario? Explain with proper **reasoning**. [CO2]

5. (a) Why do we keep a middle class for interaction between two classes sometimes? Why do we remove the middle class at other times? [CO2] (2)

(b) **Refactor** the following code: [CO2] (4)

```
public class A {
    int x;

    A(int x){
        this.x = x;
        int y = 0;
        for (int i = 2; i <= x; i++) {
            int z = 0;
            for (int j = 1; j <= i; j++) {
                if (i % j == 0) z++;
            }
            if (z == 2) y = y + i;
        }
        System.out.println("Sum of primes: " + y);
    }
}
```

```
public class B {
    int a;
    int b;

    B(int a, int b) {
        this.a = a;
        this.b = b;
        int c = a / b;
        System.out.println("Result of division: " + c);
    }
}
```